

Cutaneous Hyperpigmentation

Numerous antineoplastic drugs may cause various forms of hyperpigmentation, including patchy, acral, photoaccentuated, polycyclic, occlusion-accentuated, pressure-accentuated, or generalized patterns. The exact cause of chemotherapy-induced cutaneous hyperpigmentation remains largely unknown. Proposed mechanisms include:

- Direct toxic effect on melanocytes
- Increased drug deposition due to elevated blood flow
- Local toxicity from drug concentration in sweat
- Associated endocrinologic abnormalities
- Drug-induced depletion of tyrosinase inhibitors

Photosensitivity

Most chemotherapy-induced photosensitivity reactions are **phototoxic** rather than photoallergic in nature.

Phototoxic reactions typically develop rapidly after drug administration and resemble sunburn, presenting with burning, erythema, and sometimes vesicles or bullae. In contrast, photoallergic reactions manifest as an eczematous dermatitis and require prior sensitization, often appearing weeks to months after initial exposure.

Radiation Recall Reaction

Radiation recall reactions present as erythema strictly confined to previously irradiated skin or areas of prior sunburn. Severe cases may progress to vesiculation. True radiation recall can occur years after the original radiation exposure, and its pathogenesis is not fully understood. One theory suggests that cytotoxic agents further damage radiation-injured keratinocytes.

Methotrexate is notably associated with a unique sunburn recall reaction, typically occurring 1–3 days after UV exposure, though cases have been

reported up to 8 days later. This reaction is not prevented by folinic acid (leucovorin) supplementation.

Radiation Enhancement Reactions

Radiation enhancement refers to a synergistic interaction between an antineoplastic agent and radiation therapy, resulting in amplified skin toxicity within the irradiated field.